

MUTHU VARSHITH

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PROFESSIONAL SUMMARY

Computer Science graduate building Python-based AI/ML systems across machine learning, deep learning, NLP, multimodal AI, LLM applications and AI agents. Built and evaluated end-to-end projects with PyTorch, scikit-learn, Hugging Face Transformers, FastAPI, Flask, Docker and pytest. Focused on reliable, explainable AI systems backed by measurable evaluation and automated testing.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, HTML, CSS, Object-Oriented Programming (OOP)

AI / ML: Machine Learning, Deep Learning, Computer Vision, NLP, Model Fine-tuning, Model Evaluation, PyTorch, scikit-learn, Hugging Face Transformers

Generative AI: LLMs, LLM APIs, Prompt Engineering, Embeddings, Semantic Similarity, AI Agents

Backend / APIs: FastAPI, Flask, REST APIs, SQLAlchemy, Webhooks

Databases: PostgreSQL, SQLite

Data Science: NumPy, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis, Statistics & Probability

Developer Tools: Git, GitHub, Docker, pytest, Jupyter Notebook, VS Code

PROJECTS

Cerberus — Autonomous Software Repair & Patch Verification

Aug 2026 – Sep 2026

- Built a Python-based verification harness that independently checks whether a developer- or AI-generated patch actually fixes the reported bug before it is accepted.
- Designed an 11-stage verification pipeline that reproduces failures, applies candidate patches, reruns the test suite, and detects regressions inside network-isolated Docker sandboxes; integrated with a FastAPI-based GitHub App that posts Check Runs.
- Reduced wrong-patch approvals from 65% (13/20) to 36% (4/11) while admitting all 7 correct fixes; achieved 11/11 expected verdicts across three real open-source projects, supported by 231 automated tests.

HireAI — Explainable AI Resume Screening Platform

Jul 2026 – Aug 2026

- Developed an explainable resume-screening system that scores candidates from 0 to 100 against job requirements using semantic similarity and deterministic skill matching.
- Implemented an offline-first scoring engine that weights Sentence Transformers embeddings, fuzzy skill matching, experience and education, with Gemini 2.0 Flash as an optional LLM layer for resume parsing and explanations.
- Delivered a Flask REST API and recruiter dashboard on SQLAlchemy and SQLite with duplicate-candidate detection, skill-gap analysis, pipeline tracking and CSV/JSON export, covered by 33 pytest tests.

FusionTrace — Multimodal Deepfake Detection Platform

Mar 2025 – Jan 2026

- Fine-tuned EfficientNetV2-L on a balanced 150K-image face dataset and Wav2Vec2 on ~40K audio clips in PyTorch and Hugging Face Transformers to detect manipulated images and synthetic speech.
- Built a privacy-first multimodal pipeline that runs image, audio and video inference locally, fuses model outputs into a single risk rating and serves results through an async FastAPI backend with Grad-CAM explanations.
- Evaluated each model separately, reaching 98.3% accuracy and 0.98 F1 on a 4,002-clip held-out audio test set and 96% accuracy for the image model on its own test set.

WORK EXPERIENCE

Python Full Stack Development Intern | KodNest

Feb 2026 – May 2026

- Developed a Python-based inventory, sales and purchase management system covering stock management, multi-item invoicing, returns and payments, with role-based access control for admin, sales and inventory teams.
- Implemented stock and billing logic in atomic database transactions to enforce non-negative inventory, and validated the business rules with 59 automated tests covering overselling, returns and access control.

Artificial Intelligence Intern | SmartEd Solutions

Jun 2025 – Oct 2025

- Trained and compared Logistic Regression and SVM classifiers in scikit-learn on the 569-sample Breast Cancer Wisconsin (Diagnostic) dataset after exploratory data analysis in Pandas and Seaborn and feature standardization.
- Achieved 97.4% held-out accuracy with Logistic Regression versus 95.6% with SVM on an 80/20 train-test split, evaluated with confusion matrices and ROC curves.

EDUCATION

Visvesvaraya Technological University — B.E. in Computer Science and Business Systems

2022 – 2026

CERTIFICATIONS

AWS Prompt Engineer — Amazon Web Services, 2025

Oracle Cloud Infrastructure (OCI) 2025 Certified Generative AI Professional — Oracle, 2025

Generative AI in Action — IBM SkillsBuild, 2025

Applied AI: Practical Implementations — TechSaksham, 2025